SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1705

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3 Duration : 1 Hour Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

I M. Mounika (192411083) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mounika

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells) • Each

movement has a cost of 1, but entering risky zones has an additional cost of 2 • The robot must minimize total cost while ensuring safe navigation

• The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints

Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Hospital Doctor Scheduling Using Backtracking Search

Question

A hospital needs to assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) under the following constraints:

- D1 cannot work the Night shift.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor can be assigned to each shift.
- D3 cannot work the Morning shift.
- Every doctor must be assigned exactly one shift.

Apply the **Backtracking Search Algorithm** to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule.

Answer

Step 1: Initial State

Doctors:

- D1
- D2
- D3

Available Shifts:

- Morning
- Afternoon
- Night

Initially, no doctor is assigned to any shift.

Step 2: Assign Shift to D1

Possible shifts for D1:

- Morning ✓
- Afternoon ✓
- Night ✗ (Not allowed because D1 cannot work Night.)

Assign **Morning** to D1.

Current Assignment:

D1 → Morning

Constraint Check:

- D1 is not assigned to Night. ✓
- No shift is assigned to more than one doctor. ✓

Step 3: Assign Shift to D2

Remaining shifts:

- Afternoon
- Night

Try assigning **Afternoon** to D2.

Current Assignment:

D1 → Morning

D2 → Afternoon

Constraint Check:

- One doctor per shift. ✓
- D2 must work before D3 (to be verified after assigning D3).

Step 4: Assign Shift to D3

Remaining shift:

- Night

Assign Night to D3.

Current Assignment:

D1 → Morning

D2 → Afternoon

D3 → Night

Constraint Check:

- D3 is not assigned Morning. ✓
- D2 works before D3 (Afternoon comes before Night). ✓
- Each doctor has exactly one shift. ✓
- Only one doctor is assigned to each shift. ✓

All constraints are satisfied.

Backtracking Explanation

If an invalid assignment had been selected (for example, assigning D2 to Night), D3 would be forced to take Afternoon or Morning. Since D3 cannot work Morning and D2 must work before D3, the constraints would fail. The algorithm would backtrack to the previous assignment and try another available shift.

Final Valid Schedule

Shift	**Doctor**
Morning	D1
Afternoon	D2
Night	D3

Result

Using the **Backtracking Search Algorithm**, a valid schedule is obtained after checking all constraints. The final assignment satisfies every scheduling requirement without violating any constraints.

2. Robot Navigation Using BFS Search Algorithm

Question

A robot operates in a **5 × 5 grid**. It must move from the Start (S) position to the Goal (G). Obstacles block some paths.

Constraints:

- Movement is allowed only in Up, Down, Left, and Right directions.
- Each move has a cost of 1.
- The robot cannot move through obstacles.
- Manhattan Distance is used as the heuristic.

Use the **Breadth-First Search (BFS)** algorithm and show the node expansion, queue updates, optimal path, and total cost.

Answer

Grid Representation

S . . X .

. X . X .

. X . . .

. . X X .

X . . . G

Where:

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

Start Position = (0,0)

Goal Position = (4,4)

Manhattan Distance

The Manhattan Distance is calculated as:

h(n) = |Goal_x − Current_x| + |Goal_y − Current_y|

It estimates the number of moves required to reach the goal.

FS Expansion (Queue Updates)

- Start queue: [(0,0)]
- Expand (0,0) → add (0,1), (1,0)
- Expand (0,1) → add (0,2)
- Expand (1,0) → add (2,0)
- Expand (0,2) → add (1,2)
- Expand (2,0) → add (3,0)
- Expand (1,2) → add (2,2)
- Expand (3,0) → add (3,1)

- Expand (2,2) → add (2,3)
- Expand (2,3) → add (2,4)
- Expand (2,4) → add (3,4)
- Expand (3,4) → add (4,4) → Goal reached.

Optimal Path

(0,0) → (1,0) → (2,0) → (3,0) → (3,1)

↓

(4,1) → (4,2) → (4,3) → (4,4)

Cost Calculation

Each move costs **1**.

Number of moves = **8**

Total Cost = **8** × **1 = 8**

Result

The Breadth-First Search algorithm successfully finds the shortest path from the Start node to the Goal node while avoiding obstacles. The optimal path requires **8 moves**, resulting in a total path cost of **8**.

3. Autonomous Rescue Robot in a Disaster Environment

Question

An autonomous rescue robot is deployed in a disaster-affected building to locate survivors. The robot operates in a partially observable environment where some paths are blocked and risky areas increase travel cost.

Answer the following:

a) Analyze the constraints.

b) Develop an appropriate search strategy.

c) Explain Intelligent Agent, Rationality, and Environment Type.

d) Justify why the chosen approach is the best.

Answer

(a) Constraint Analysis

The rescue robot must satisfy several important constraints while navigating through the disaster area.

Movement Constraint

The robot is allowed to move only in four directions:

- Up
- Down
- Left
- Right

This limits the available paths and simplifies movement decisions.

Obstacle Constraint

Some rooms contain obstacles.

The robot cannot move through these blocked cells and must find an alternative safe route.

Limited Observation

The robot can observe only the adjacent cells.

It does not know the complete map initially and must gradually discover the environment while moving.

Movement Cost

Every movement has a basic cost of **1**.

The robot should minimize unnecessary movements.

Risk Zone Constraint

Moving into risky areas increases the movement cost by an additional **2**.

Therefore:

- Normal Cell Cost = 1
- Risky Cell Cost = 3

The robot should avoid risky areas whenever possible.

Dynamic Knowledge

As the robot explores the building, it continuously updates its internal map using newly discovered information.

If a blocked path is detected, the robot immediately replans its route.

(b) Solution Approach Using Uniform Cost Search (UCS) and Online Search

The most suitable solution is to combine **Uniform Cost Search (UCS)** with an **Online Search Agent**.

Step 1

The robot starts from the initial position **S**.

Step 2

It senses all neighboring cells.

Unknown areas become known only after observation.

Step 3

The robot assigns movement costs:

- Normal cell = 1
- Risky cell = 3

Step 4

Uniform Cost Search stores all possible paths in a priority queue.

The path with the smallest cumulative cost is selected first.

Step 5

Whenever new obstacles are discovered, the robot updates its map and recalculates the safest and least costly route.

Step 6

This process continues until the robot reaches the survivor located at **Goal (G)**.

Advantages

- Finds the minimum-cost path.
- Avoids risky areas whenever possible.
- Adapts to unknown environments.
- Continuously updates decisions based on new information.

(c) AI Concepts

Intelligent Agent

The rescue robot is an intelligent agent because it:

- Collects information using sensors.
- Processes environmental data.
- Makes decisions independently.
- Performs actions through motors.
- Learns new information while exploring.

Rationality

A rational agent always selects the action that produces the best outcome.

The rescue robot behaves rationally by:

- Choosing the safest route.
- Minimizing total travel cost.
- Avoiding blocked paths.
- Replanning whenever new obstacles are detected.

Environment Type

The rescue environment can be classified as follows:

Partially Observable

The robot cannot view the entire building at once.

Single Agent

Only one robot performs the rescue task.

Dynamic Environment

The robot's knowledge changes while exploring.

Discrete Environment

The robot moves from one grid cell to another.

Deterministic Environment

Each action produces a predictable movement and cost.

(d) Justification

Uniform Cost Search combined with an Online Search Agent is the most suitable approach because:

- It guarantees the minimum-cost path.
- It considers both movement costs and additional risk costs.
- It can adapt to newly discovered obstacles in a partially observable environment.

- It supports dynamic replanning during navigation.
- It ensures efficient and safe rescue operations.

Compared to Breadth-First Search, Uniform Cost Search is more appropriate because BFS assumes equal path costs, whereas UCS correctly handles different movement costs caused by risky zones.

Conclusion

The given problems demonstrate different Artificial Intelligence problem-solving techniques.

- **Backtracking Search** efficiently solves the hospital scheduling problem by checking constraints and backtracking whenever a conflict occurs.
- **Breadth-First Search (BFS)** finds the shortest path in a grid with equal movement costs while avoiding obstacles.
- **Uniform Cost Search (UCS)** with an **Online Search Agent** is the best choice for autonomous rescue robots because it minimizes total travel cost, avoids risky areas, and dynamically updates its decisions in partially observable environments.

These AI techniques help solve real-world scheduling, path-finding, and decision-making problems efficiently while satisfying all given constraints.